EXDA Metadata Report

Project: VH2D-Project

Generated: 2026-05-26 06:44:28

Includes: Plan, DAQ Systems, Sensors Mapping, Gas Mixing

Responsible Researcher: PhD Student Javier I. Camacho

Project Objective:

LU / DBI / Kistler pressure DAQ systems characterisation.

Project Description:

The primary objectives of the experiments are:

1. Internal Over-pressure Development

Measurement of internal over-pressure development during vented hydrogen deflagrations in a partially confined enclosure for homogeneous flammable hydrogen·air mixtures with hydrogen concentrations up to 19 vol.%.

2.Flame Propagation (Under Consideration)

Measurement of flame propagation characteristics, including potential estimation of flame time-of-arrival (TOA) at selected locations.

3.Instrumentation Evaluation

Evaluation and cross-comparison of different pressure measurement systems and data acquisition configurations, including sensor characteristics and the influence of measurement chain dynamics (e.g., time constant (tau) effects, sampling rate).

4. Measurement Uncertainty Factors

Investigation of experimental factors that may affect the accuracy and reliability of pressure measurements, including structural vibration effects and thermal protection methods.

5. Venting Behaviour

Investigation of vent opening behaviour and the influence of vent rupture mechanisms on pressure development, including comparison between pressure-driven thin-film rupture and controlled vent opening using a hot-wire cutting system.

EXDA Plan Export - Experiment_Plan_v000

Responsible Researcher: PhD Student Javier I. Camacho
Objective: LU / DBI / Kistler pressure DAQ systems characterisation.
Start: 2026-05-25 | Deadline: 2026-05-29
Total Experiments: 24
Generated: 2026-05-26 06:44:28

Group: Prep
Group Objective: Preparation

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
Prep-01	Yes	2026-05-25	13.5	back	Thin Plas·	101325	292.15	3	Prep-01/DAQ-1/Prep-01.exp/data/Prep_01-DAQ-1.tpc5
									Prep-01/DAQ-2/Prep-01-DAQ-2.mf4
									Prep-01/H2CM/Prep-01-H2CM-U.csv

Group: Verif
Group Objective: Performance Verification

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
Verif-01	No	2026-05-26	12			101325	293	0	-
Verif-02	No	2026-05-26	14			101325	293	0	-
Verif-03	No	2026-05-26	16			101325	293	0	-
Verif-04	No	2026-05-26	18			101325	293	0	-

Group: VH2D-01
Group Objective: Vented deflagration concentration series (Hotwire)

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-01-01	No	2026-05-26	12	Back	Thin Plas·	101325	293	0	-
VH2D-01-02	No	2026-05-26	12			101325	293	0	-
VH2D-01-03	No	2026-05-26	14			101325	293	0	-
VH2D-01-04	No	2026-05-26	14			101325	293	0	-
VH2D-01-05	No	2026-05-26	16			101325	293	0	-
VH2D-01-06	No	2026-05-26	16			101325	293	0	-
VH2D-01-07	No	2026-05-26	18			101325	293	0	-
VH2D-01-08	No	2026-05-26	18			101325	293	0	-

Group: VH2D-02
Group Objective: Vented deflagration concentration series (NO Hotwire)

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-02-01	No	2026-05-26	12			101325	293	0	-
VH2D-02-02	No	2026-05-26	14			101325	293	0	-
VH2D-02-03	No	2026-05-26	16			101325	293	0	-
VH2D-02-04	No	2026-05-26	18			101325	293	0	-

Group: VH2D-03

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-03-01	No	2026-05-27	12			101325	293	0	-
VH2D-03-02	No	2026-05-27	14			101325	293	0	-
VH2D-03-03	No	2026-05-27	16			101325	293	0	-
VH2D-03-04	No	2026-05-27	18			101325	293	0	-

Group: VH2D-04

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-04-01	No	2026-05-28	18			101325	293	0	-

Group: VH2D-05

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-05-01	No	2026-05-29				101325	293	0	-

Group: VH2D-06

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-06-01	No	2026-05-29				101325	293	0	-

EXDA DAQ Systems Export

Project: VH2D-Project
Responsible Researcher: PhD Student Javier I. Camacho
Total DAQ Systems: 5
Reference PDF: LU-DBI Measurement Chain (frontend/public/LU-DBI_MeasurementChain_v001.pdf)
Reference PDF: Mixture Sampling Sub-System (frontend/public/MixtureSampling_Sub-System_000.pdf)
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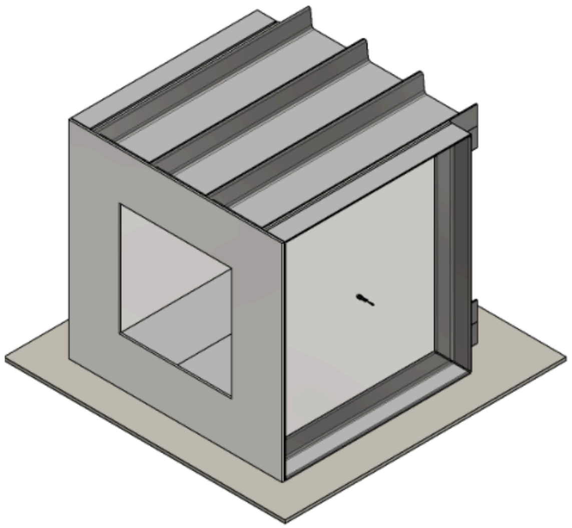
Name	Quantity	Vendor	Model	Serial	Fs (Hz)	Ch	Owner	Last Cal.	Cal Cert ID	Active	Notes
DAQ-1	pressure	Elsys - Kistler	TraNET 204S8	441309	-	8	LU	2024-10-21	-	Yes	2 equipement
DAQ-2	pressure	Kistler	5165A4	6306100	-	4	DBI	-	-	Yes	-
DAQ-3	pressure	Kistler	5165A4	5358310	-	4	Kistler	-	-	Yes	-
DAQ-4	pressure	Kistler	-	-	-	8	Kistler	-	-	Yes	-
H2CM	concentration	JCT	JTCD-300	14362	1	-	LU	2025-07-21	-	Yes	-

DAQ Reference Diagram - LU-DBI Measurement Chain

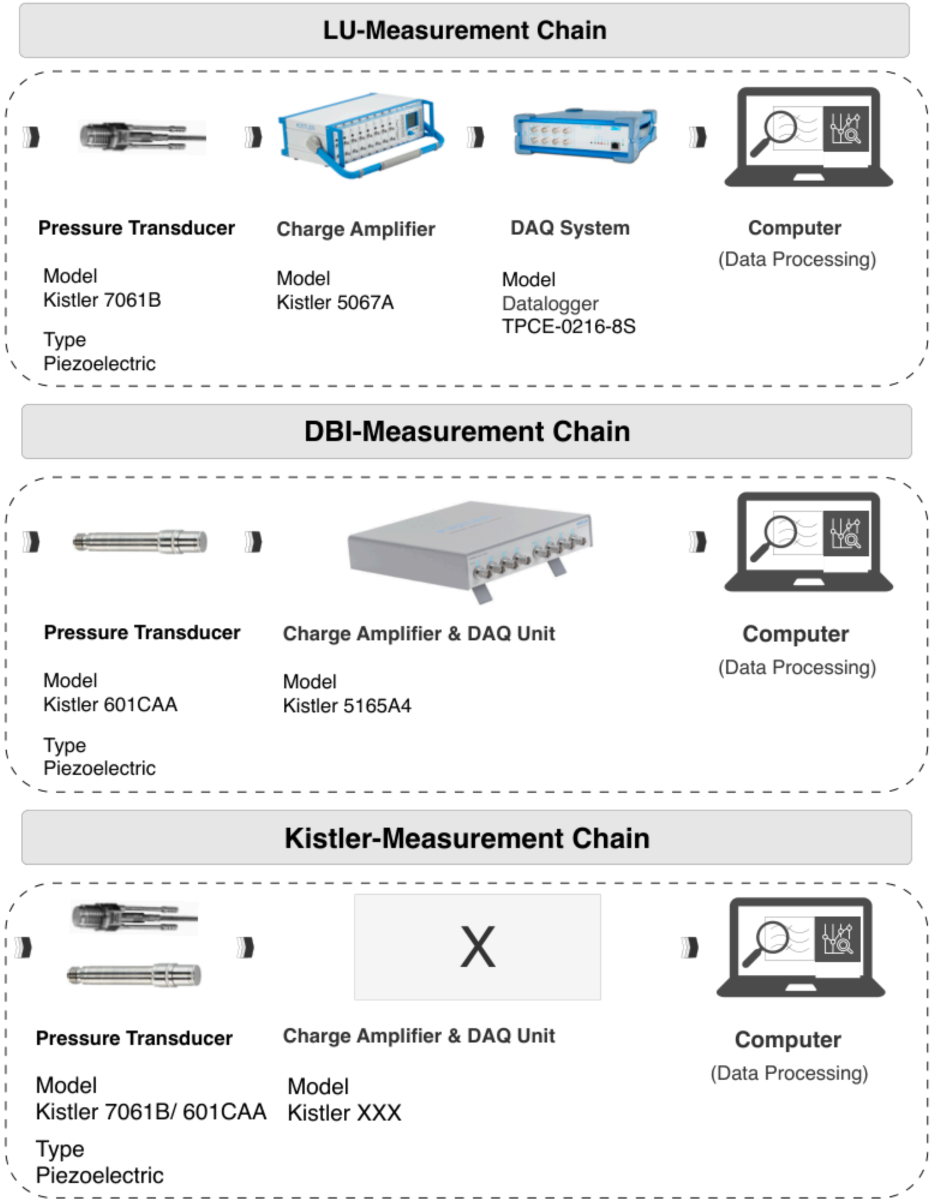
Source: frontend/public/LU-DBI_MeasurementChain_v001.pdf



Explosion Chamber

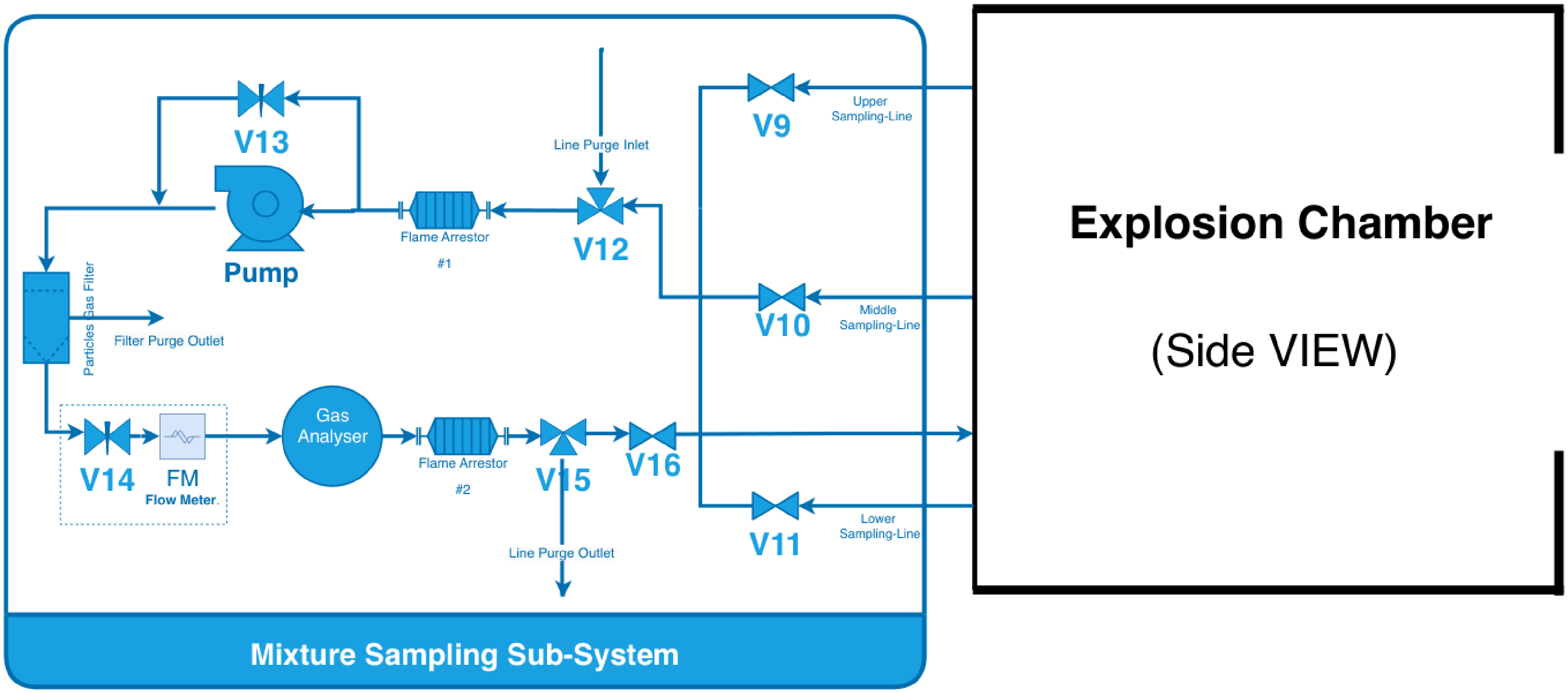
Dimensions (Internal): 900 x 900 x 900 [mm]

Material: Steel



DAQ Reference Diagram - Mixture Sampling Sub-System

Source: frontend/public/MixtureSampling_Sub-System_000.pdf



EXDA Sensors Mapping Export

Project: VH2D-Project
Responsible Researcher: PhD Student Javier I. Camacho
Total Groups: 7
Total Sensor IDs: 11
Total Sensor Mappings: 15
Sensor Mounting Reference File: frontend/public/SensorMountingLocation-000.pdf
Generated: 2026-05-26 06:44:28

Group: Prep

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi·	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
Prep	P1	pressure	DAQ-1	CH1	1254024	-80.88 pC/bar	5L	(0.45,0.593,0.396)	flush	Yes	No	-	Complete
Prep	P2	pressure	DAQ-2	CH2	6266801	-37.02 pC/bar	6L	(0.450,0.290,0.403)	flush	Yes	No	-	Complete
Prep	Trig-1	voltage	DAQ-1	CH5	-	- pC/bar	-	(-,-,-)	N/A	Yes	No	M-Duino	Complete
Prep	Trig-2	voltage	DAQ-2	CH4	-	- pC/bar	-	(-,-,-)	N/A	Yes	No	M-Duino	Complete

Group: VH2D-01

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi·	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-01	P1	pressure	DAQ-1	CH1	1254024	-80.88 pC/bar	5L	(0.45,0.593,0.396)	flush	Yes	No	-	Complete
VH2D-01	P2	pressure	DAQ-2	CH1	6266801	-37.02 pC/bar	6L	(0.450,0.290,0.403)	flush	Yes	No	-	Complete
VH2D-01	P3	pressure	DAQ-1	CH2	-	- pC/bar	-	(-,-,-)	other	Yes	Yes	-	Incomplete
VH2D-01	P4	pressure	DAQ-2	CH2	-	- pC/bar	-	(-,-,-)	flush	Yes	No	-	Incomplete
VH2D-01	P5	pressure	DAQ-1	CH3	-	- pC/bar	-	(-,-,-)	flush	Yes	No	-	Incomplete
VH2D-01	P6	pressure	DAQ-2	CH3	-	- pC/bar	-	(-,-,-)	flush	Yes	No	-	Incomplete
VH2D-01	P7	pressure	DAQ-2	CH3	-	- pC/bar	-	(-,-,-)	flush	Yes	No	-	Incomplete
VH2D-01	P8	pressure	DAQ-3	CH1	-	- pC/bar	-	(-,-,-)	flush	Yes	No	-	Incomplete
VH2D-01	Trig-1	voltage	DAQ-1	CH5	-	- pC/bar	-	(-,-,-)	N/A	Yes	No	M-Duino	Complete
VH2D-01	Trig-2	voltage	DAQ-2	CH4	-	- pC/bar	-	(-,-,-)	N/A	Yes	No	M-Duino	Complete
VH2D-01	Trig-3	voltage	DAQ-3	CH2	-	- pC/bar	-	(-,-,-)	N/A	Yes	No	M-Duino	Complete

Group: VH2D-02

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi·	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-02	-	-	-	-	-	-	-	-	-	-	-	-	Reference

Group Reference Note: same configuration as preparation group

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Group: VH2D-03

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi·	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-03	-	-	-	-	-	-	-	-	-	-	-	-	Reference

Group Reference Note: same configuration as preparation group

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Group: VH2D-04

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi·	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-04	-	-	-	-	-	-	-	-	-	-	-	-	Reference

Group Reference Note: same configuration as preparation group

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Group: VH2D-05

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi·	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-05	-	-	-	-	-	-	-	-	-	-	-	-	Reference

Group Reference Note: same configuration as preparation group

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Group: VH2D-06

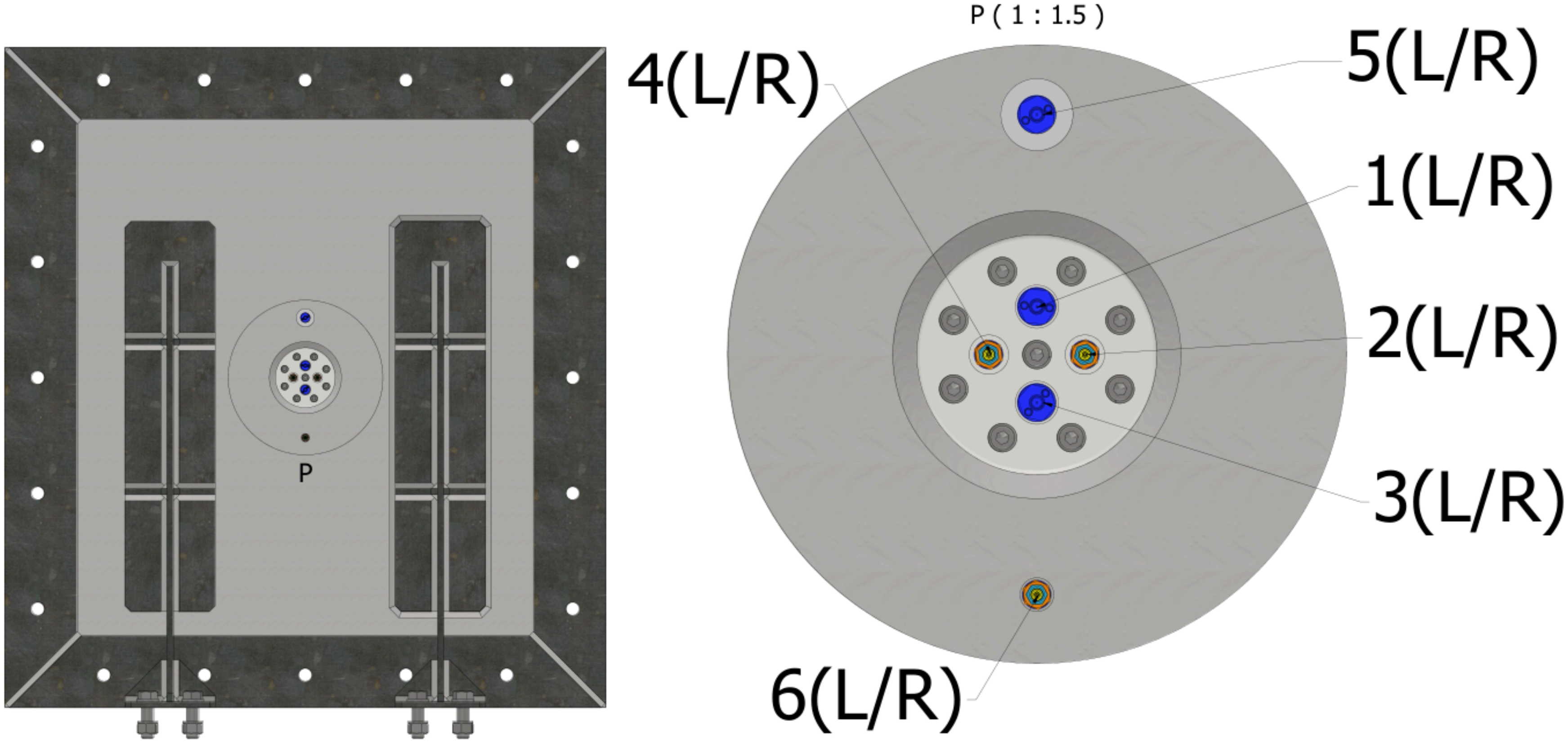
Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi·	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-06	-	-	-	-	-	-	-	-	-	-	-	-	Reference

Group Reference Note: same configuration as preparation group

Sensor Mounting Reference Diagram

Source: frontend/public/SensorMountingLocation-000.pdf

Pressure Sensor Location Mapping Overview



EXDA Gas Mixing Export

Project: VH2D-Project
Responsible Researcher: PhD Student Javier I. Camacho
Total Records: 1
Generated: 2026-05-26 06:44:28
MATLAB Verified: Yes
Verification File A: scripts/GasMixingVerificationFiles/H2_MFC_Fill_Calculator_v000.m
Verification File B: scripts/GasMixingVerificationFiles/AuxFcn_H2_MFC_FillCalculator_000.m

Shared Chamber Setup (applies to all runs unless noted otherwise):
L=0.900 m, W=0.900 m, H=0.900 m, Pipes+=0.000 m^3, Hotwire+=0.000 m^3, Welded-=0.000 m^3, Bolts-=0.000 m^3, Vcorr=729.00 L

Core Gas Mixing Results

Group	Run/Test	H2%	Pch(Pa)	Tch(K)	H2(g)	H2inj(L)	MFC	t(s)	t(min)	Updated
Prep	Prep-01	13.50	101325	292.15	8.276	98.41	20.00	301.3	5.0	2026-05-25 17:27:49